

APM1205: Linear Models

Formative Assessment 3

Bonn Andrei M. Buenaventura

09-17-2026

1. Unbiasedness of Least-Squares Estimators A and B

(a) Expected value of the slope estimator B

The least-squares slope estimator B can be expressed as a linear function of the observations: $B = \sum m_i Y_i$. The constant weights m_i are defined as:

$$m_i = \frac{x_i - \bar{x}}{\sum (x_j - \bar{x})^2} \quad (1)$$

Before computing the expectation, we establish two fundamental properties of m_i :

1. $\sum m_i = \frac{\sum (x_i - \bar{x})}{\sum (x_j - \bar{x})^2} = 0$
2. $\sum m_i x_i = \frac{\sum x_i (x_i - \bar{x})}{\sum (x_j - \bar{x})^2} = \frac{\sum (x_i - \bar{x})^2}{\sum (x_j - \bar{x})^2} = 1$

Using the assumption of linearity, $E(Y_i) = \alpha + \beta x_i$, we take the expectation of B :

$$\begin{aligned} E(B) &= E\left(\sum m_i Y_i\right) \\ &= \sum m_i E(Y_i) \\ &= \sum m_i (\alpha + \beta x_i) \\ &= \alpha \sum m_i + \beta \sum m_i x_i \end{aligned}$$

Substituting the established properties of m_i :

$$E(B) = \alpha(0) + \beta(1) = \beta \quad (2)$$

This confirms that B is an unbiased estimator of the population slope β .

(b) Expected value of the intercept estimator A

The least-squares intercept estimator A is given by $A = \bar{Y} - B\bar{x}$. By substituting $\bar{Y} = \frac{1}{n} \sum Y_i$ and $B = \sum m_i Y_i$, we can rewrite A as a linear combination of the Y_i observations:

$$\begin{aligned} A &= \frac{1}{n} \sum Y_i - \bar{x} \sum m_i Y_i \\ &= \sum \left(\frac{1}{n} - \bar{x} m_i \right) Y_i \end{aligned}$$

Let $c_i = \left(\frac{1}{n} - \bar{x}m_i\right)$. Taking the expectation of A :

$$\begin{aligned} E(A) &= \sum c_i E(Y_i) \\ &= \sum \left(\frac{1}{n} - \bar{x}m_i\right) (\alpha + \beta x_i) \\ &= \alpha \sum \left(\frac{1}{n} - \bar{x}m_i\right) + \beta \sum \left(\frac{1}{n} - \bar{x}m_i\right) x_i \end{aligned}$$

We evaluate the two sums independently:

1. $\sum \left(\frac{1}{n} - \bar{x}m_i\right) = \sum \frac{1}{n} - \bar{x} \sum m_i = 1 - \bar{x}(0) = 1$
2. $\sum \left(\frac{1}{n} - \bar{x}m_i\right) x_i = \frac{1}{n} \sum x_i - \bar{x} \sum m_i x_i = \bar{x} - \bar{x}(1) = 0$

Substituting these back into the expectation equation yields:

$$E(A) = \alpha(1) + \beta(0) = \alpha \quad (3)$$

This demonstrates that A is an unbiased estimator of the population intercept α .

2. Sampling Variances of A and B

To derive the variances, we apply the Gauss-Markov assumptions: linearity, constant variance where $V(Y_i) = \sigma_\varepsilon^2$, and independence where the covariance between any Y_i and Y_j is zero.

Variance of B:

Starting from the linear form $B = \sum m_i Y_i$, and because the observations are independent, the variance of the sum is the sum of the variances:

$$V(B) = V\left(\sum m_i Y_i\right) = \sum m_i^2 V(Y_i) = \sigma_\varepsilon^2 \sum m_i^2 \quad (4)$$

Evaluating $\sum m_i^2$:

$$\sum m_i^2 = \sum \left(\frac{x_i - \bar{x}}{\sum (x_j - \bar{x})^2}\right)^2 = \frac{\sum (x_i - \bar{x})^2}{[\sum (x_j - \bar{x})^2]^2} = \frac{1}{\sum (x_i - \bar{x})^2} \quad (5)$$

Substituting this back yields the variance of the slope:

$$V(B) = \frac{\sigma_\varepsilon^2}{\sum (x_i - \bar{x})^2} \quad (6)$$

Variance of A:

Using the linear form $A = \sum c_i Y_i$ where $c_i = \left(\frac{1}{n} - \bar{x}m_i\right)$:

$$V(A) = \sum c_i^2 V(Y_i) = \sigma_\varepsilon^2 \sum \left(\frac{1}{n} - \bar{x}m_i\right)^2 \quad (7)$$

Expanding the squared term:

$$\begin{aligned} \sum \left(\frac{1}{n} - \bar{x}m_i\right)^2 &= \sum \left(\frac{1}{n^2} - \frac{2\bar{x}m_i}{n} + \bar{x}^2 m_i^2\right) \\ &= \frac{n}{n^2} - \frac{2\bar{x}}{n} \sum m_i + \bar{x}^2 \sum m_i^2 \\ &= \frac{1}{n} - 0 + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2} \end{aligned}$$

Substituting this sum back into the variance equation yields the variance of the intercept:

$$V(A) = \sigma_\varepsilon^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2} \right) \quad (8)$$

3. Linear Transformation of X and Y

Using the reference baseline of Davis's regression of measured on reported weight, we establish the original estimates:

- $B = 0.97722$
- $SE(B) = 0.030529$
- $t_0 = \frac{B}{SE(B)} = \frac{0.97722}{0.030529} \approx 32.01$

(a) X-values transformed by $X' = 10(X - 1)$

When X is scaled by a factor of 10, the deviations from the mean scale proportionally: $(X'_i - \bar{X}') = 10(X_i - \bar{X})$. The sum of squared deviations scales by $10^2 = 100$: $\sum (X'_i - \bar{X}')^2 = 100 \sum (X_i - \bar{X})^2$.

Calculating the new slope B' :

$$B' = \frac{\sum (X'_i - \bar{X}')(Y_i - \bar{Y})}{\sum (X'_i - \bar{X}')^2} = \frac{10 \sum (X_i - \bar{X})(Y_i - \bar{Y})}{100 \sum (X_i - \bar{X})^2} = \frac{1}{10} B = 0.097722 \quad (9)$$

Calculating the new standard error $SE(B')$:

$$SE(B') = \frac{SE}{\sqrt{\sum (X'_i - \bar{X}')^2}} = \frac{SE}{\sqrt{100 \sum (X_i - \bar{X})^2}} = \frac{1}{10} SE(B) = 0.0030529 \quad (10)$$

Calculating the transformed t -statistic t'_0 :

$$t'_0 = \frac{B'}{SE(B')} = \frac{B/10}{SE(B)/10} = \frac{B}{SE(B)} = 32.01 \quad (11)$$

(b) Y-values transformed by $Y'' = 5(Y + 2)$

When Y is scaled by a factor of 5, the deviations from the mean scale proportionally: $(Y''_i - \bar{Y}'') = 5(Y_i - \bar{Y})$.

Calculating the new slope B'' :

$$B'' = \frac{\sum (X_i - \bar{X})(Y''_i - \bar{Y}'')}{\sum (X_i - \bar{X})^2} = \frac{5 \sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2} = 5B = 4.8861 \quad (12)$$

Calculating the new standard error $SE(B'')$. The residuals themselves scale by 5, meaning $SE'' = 5SE$:

$$SE(B'') = \frac{SE''}{\sqrt{\sum (X_i - \bar{X})^2}} = \frac{5SE}{\sqrt{\sum (X_i - \bar{X})^2}} = 5SE(B) = 0.152645 \quad (13)$$

Calculating the transformed t -statistic t''_0 :

$$t''_0 = \frac{B''}{SE(B'')} = \frac{5B}{5SE(B)} = 32.01 \quad (14)$$

(c) General Impact on Inference

Linear transformations (scaling or shifting) of X and Y alter the magnitude and units of the slope estimates B and their associated standard errors $SE(B)$ in direct proportion to the scaling factors applied. However, because both the coefficient and the standard error scale by the exact same proportional constant, the scalar cancels out when computing the test statistic $t_0 = B/SE(B)$ [cite: 2, 7]. Consequently, t -statistics, corresponding p -values, and hypothesis tests for β remain completely invariant under linear transformations. Confidence intervals shift and stretch according to the scaling factor, accurately reflecting the new units without altering the underlying statistical certainty[cite: 8].